Table S4: Degrees of freedom (in thousands) of diffused Nitsche (three per free node of the trimmed octree) and of the three body-fitted P1 meshes (three per vertex) at octree depth 7. Matched DOFs: vertex count targeted at the octree's node count (secant search, closest of up to six trials kept); matched surface: surface edge length $h_{\min}$; graded: $h_{\min}$ at the surface, coarser inside like the octree. Flags, at the session's $h_{\min}$: T, wall thinner than $4h_{\min}$; G, gaps narrower than $h_{\min}$ on over 1% of the surface; L, a load within $3h_{\min}$ of a support; S, a support piece below $50h_{\min}^2$ or a partial clamp on a section thinner than $4h_{\min}$; blank, unflagged. —, no body-fitted mesh (the input surface self-intersects). Medians over all objects and over the unflagged ones; n, number of values. Details in Section S**??**.

			Body-fitted P1						Body-fitted P1		
Object	Flags	Diffused Nitsche	Matched DOFs	Matched surface	Graded	Object	Flags	Diffused Nitsche	Matched DOFs	Matched surface	Graded
a TV remote control		49.4	49.5	57.0	49.0	a car suspension control arm with rubber bushings		82.1	81.7	76.9	80.1
a USB stick		43.1	43.2	50.3	42.3	a truck leaf spring		16.0	16.0	7.00	15.2
an electric toothbrush		17.2	17.2	11.0	17.2	a car wheel with its rubber tyre	G	308	279	708	318
a coffee mug	T	260	260	249	256	a car brake disc	S	184	—	—	—
a computer mouse		108	108	237	108	a crankshaft	G	65.3	64.8	68.9	64.6
a school chair with a steel frame and a plastic seat	T, S	83.2	59.6	33.0	82.7	a road roller with a steel drum and rubber tyres	T, G, S	101	101	64.1	101
a glass-top dining table	T, S	73.8	72.7	53.4	71.7	a jackhammer	S	21.9	21.3	12.7	21.4
a park bench	T, S	94.4	—	—	—	an industrial robot arm	G	90.3	91.0	97.8	87.1
a bar stool	S	86.1	84.4	46.5	83.9	an adjustable wrench	T, S	18.7	18.3	8.36	18.8
a sofa	S	132	133	195	131	a claw hammer		27.7	27.2	18.4	27.2
a teddy bear		116	116	192	116	a dental crown		357	370	1300	364
a wooden rocking horse	T	98.4	97.1	63.0	95.7	a knee implant	G	291	296	654	287
a pogo stick	S	14.3	14.6	3.39	13.8	a walking cane with a rubber tip	T, L, S	8.27	8.42	1.99	8.09
a wooden toy train engine	G, L, S	103	104	149	102	a human femur		36.6	35.7	28.4	35.5
a yo-yo	G	339	371	761	337	a lumbar vertebra		99.6	99.3	142	99.3
a jelly dessert		200	202	609	200	a skyscraper	L	52.8	53.0	80.8	51.8
a loaf of bread		235	238	843	247	a thatched stone cottage	L	219	221	720	218
a hamburger	G	237	238	837	248	a lighthouse		99.7	91.0	160	97.0
a layered birthday cake		196	197	570	195	an igloo		245	245	292	244
a slice of watermelon		113	113	245	113	a log cabin	L	264	270	814	263
a giraffe	S	36.4	35.3	23.9	35.1	the Leaning Tower of Pisa	T, G, S	86.0	182	228	127
an elephant	S	95.8	96.3	127	95.5	the Pantheon	L	206	207	256	206
a rhinoceros	S	79.1	79.8	114	78.5	Himeji Castle	G	313	—	—	—
a horse	S	86.1	85.9	95.8	85.3	a natural stone arch	G	193	196	353	192
a crocodile	S	40.3	39.6	33.7	38.6	the Taj Mahal		209	213	422	207
Median (all objects)		99.0	97.1	127	97.0	Median (unflagged)		104	104	176	103
n		50	47	47	47	n		18	18	18	18

Table S5: Discrepancy of diffused Nitsche from each body-fitted P1 solution at octree depth 7. η_{rig} and $\eta_{\text{rig}}^{H^1}$: relative L^2 and H^1 norms of the displacement difference after removing its best-fit rigid motion (in %); $\ell_{\text{N}}/\ell_{\text{P1}}$: compliance ratio (below 1: Nitsche stiffer); s_{95}, s_{99}: Nitsche-to-P1 ratios of the 95th and 99th percentiles of the von Mises stress. Flags, at the session's $h_{\min}$: T, wall thinner than $4h_{\min}$; G, gaps narrower than $h_{\min}$ on over 1% of the surface; L, a load within $3h_{\min}$ of a support; S, a support piece below $50h_{\min}^2$ or a partial clamp on a section thinner than $4h_{\min}$; blank, unflagged. Gray rows: η_{rig} above 10% against at least one mesh. —, no body-fitted mesh (the input surface self-intersects). Medians over all objects and over the unflagged ones; n, number of values. Details in Section S**??**.

		Body-fitted P1														
		Matched DOFs					Matched surface					Graded				
Object	Flags	η_{rig} (%)	$\eta_{\text{rig}}^{H^1}$ (%)	$\ell_{\text{N}}/\ell_{\text{P1}}$	s_{95}	s_{99}	η_{rig} (%)	$\eta_{\text{rig}}^{H^1}$ (%)	$\ell_{\text{N}}/\ell_{\text{P1}}$	s_{95}	s_{99}	η_{rig} (%)	$\eta_{\text{rig}}^{H^1}$ (%)	$\ell_{\text{N}}/\ell_{\text{P1}}$	s_{95}	s_{99}
a TV remote control		3.7	7.0	0.909	0.924	0.893	2.9	5.8	0.935	0.942	0.910	2.8	5.7	0.937	0.943	0.912
a USB stick		1.4	4.4	0.871	0.886	0.841	1.1	3.8	0.886	0.904	0.847	1.0	3.6	0.886	0.907	0.851
an electric toothbrush		15	22	0.773	0.828	0.755	8.4	13	0.876	0.858	0.823	12	17	0.824	0.826	0.761
a coffee mug	T	4.7	10	0.869	0.885	0.842	4.6	10	0.872	0.887	0.840	4.8	10	0.868	0.884	0.837
a computer mouse		7.0	21	0.935	0.972	0.937	6.4	21	0.945	0.974	0.933	6.1	21	0.949	0.977	0.932
a school chair with a steel frame and a plastic seat	T, S	34	46	0.545	0.618	0.542	32	42	0.599	0.649	0.546	35	47	0.533	0.605	0.513
a glass-top dining table	T, S	16	28	0.852	0.893	0.953	12	21	0.907	0.940	1.003	15	26	0.870	0.909	0.969
a park bench	T, S	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
a bar stool	S	15	30	0.862	0.911	0.818	12	27	0.865	0.901	0.831	15	31	0.845	0.888	0.803
a sofa	S	1.9	9.2	0.937	0.908	0.920	2.1	9.7	0.939	0.904	0.918	2.0	9.6	0.943	0.912	0.911
a teddy bear		1.5	4.2	0.831	0.878	0.893	1.6	4.3	0.837	0.927	0.877	1.6	4.3	0.839	0.928	0.876
a wooden rocking horse	T	9.1	16	0.857	0.883	0.888	8.9	15	0.857	0.887	0.881	10	17	0.844	0.873	0.873
a pogo stick	S	20	71	0.864	0.782	1.068	9.5	35	0.881	0.766	1.017	9.1	35	0.845	0.732	1.014
a wooden toy train engine	G, L, S	8.4	42	0.883	0.950	0.823	8.4	42	0.886	0.952	0.822	8.0	42	0.886	0.949	0.825
a yo-yo	G	21	61	0.461	0.676	0.565	21	62	0.454	0.669	0.554	21	61	0.460	0.673	0.561
a jelly dessert		2.7	23	0.957	0.958	0.963	3.2	22	0.963	0.948	0.960	2.6	23	0.969	0.962	0.974
a loaf of bread		4.8	23	0.886	0.924	0.863	4.5	23	0.902	0.938	0.877	4.2	23	0.907	0.944	0.879
a hamburger	G	7.1	23	0.911	0.937	0.859	7.6	24	0.908	0.927	0.843	5.8	24	0.930	0.944	0.869
a layered birthday cake		3.0	19	0.976	0.969	0.979	2.8	20	0.985	0.961	0.978	2.4	21	0.989	0.962	0.977
a slice of watermelon		4.0	21	0.923	0.951	0.897	3.7	20	0.926	0.949	0.886	3.4	20	0.931	0.953	0.889
a giraffe	S	2.7	7.5	1.016	0.946	1.180	2.1	6.1	1.030	0.939	1.146	2.5	7.1	1.019	0.932	1.152
an elephant	S	1.1	4.1	0.981	0.974	0.981	1.2	4.2	0.984	0.979	0.984	1.2	4.2	0.984	0.976	0.986
a rhinoceros	S	2.2	8.3	0.945	0.940	0.976	2.3	8.4	0.952	0.947	0.976	2.3	8.4	0.954	0.954	0.977
a horse	S	28	62	0.861	0.915	0.942	27	59	0.859	0.909	0.942	27	59	0.848	0.910	0.942
a crocodile	S	8.0	16	0.634	0.759	0.602	8.0	16	0.637	0.747	0.598	8.1	17	0.627	0.741	0.608
a car suspension control arm with rubber bushings		7.5	23	0.912	0.934	0.937	4.6	16	0.893	0.916	0.909	4.6	16	0.891	0.915	0.908
a truck leaf spring		24	37	0.710	0.693	0.642	17	27	0.793	0.742	0.727	22	33	0.736	0.689	0.672
a car wheel with its rubber tyre	G	7.4	16	0.822	0.901	0.816	8.3	18	0.799	0.882	0.795	7.7	17	0.816	0.897	0.811
a car brake disc	S	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
a crankshaft	G	12	31	0.469	0.511	0.445	12	30	0.494	0.532	0.475	12	30	0.494	0.532	0.469
a road roller with a steel drum and rubber tyres	T, G, S	51	57	0.372	0.550	0.807	47	54	0.399	0.571	0.841	49	56	0.384	0.558	0.825
a jackhammer	S	6.6	24	0.547	0.856	0.539	11	32	0.605	0.821	0.542	13	38	0.549	0.814	0.494
an industrial robot arm	G	23	54	0.424	0.649	0.573	23	54	0.417	0.640	0.564	23	54	0.418	0.641	0.565
an adjustable wrench	T, S	2.1	7.1	1.077	1.099	1.092	2.6	8.8	1.094	1.112	1.105	2.5	8.7	1.083	1.103	1.091

Continued on next page

Table S5: Discrepancy at octree depth 7 (continued).

		Body-fitted P1														
		Matched DOFs					Matched surface					Graded				
Object	Flags	η_{rig} (%)	$\eta_{\text{rig}}^{H^1}$ (%)	$\ell_{\text{N}}/\ell_{\text{P1}}$	s_{95}	s_{99}	η_{rig} (%)	$\eta_{\text{rig}}^{H^1}$ (%)	$\ell_{\text{N}}/\ell_{\text{P1}}$	s_{95}	s_{99}	η_{rig} (%)	$\eta_{\text{rig}}^{H^1}$ (%)	$\ell_{\text{N}}/\ell_{\text{P1}}$	s_{95}	s_{99}
a claw hammer		1.6	5.9	0.879	0.874	0.868	1.8	6.5	0.872	0.866	0.855	2.0	7.1	0.862	0.860	0.844
a dental crown		7.7	24	0.943	0.866	0.811	7.7	25	0.945	0.860	0.807	7.5	25	0.946	0.864	0.808
a knee implant	G	80	99	0.403	0.437	0.340	80	99	0.398	0.432	0.337	80	99	0.406	0.438	0.341
a walking cane with a rubber tip	T, L, S	24	39	0.921	0.807	0.741	18	30	0.938	0.862	0.741	21	35	0.925	0.837	0.703
a human femur		0.64	2.2	0.974	0.973	0.963	1.2	3.2	0.956	0.955	0.953	1.4	3.7	0.948	0.941	0.936
a lumbar vertebra		6.9	23	0.933	0.898	0.885	6.7	23	0.946	0.902	0.889	6.6	23	0.946	0.903	0.890
a skyscraper	L	0.35	0.37	0.992	0.981	0.979	0.34	0.37	1.004	0.980	0.983	0.32	0.34	1.003	0.981	0.982
a thatched stone cottage	L	0.73	4.9	0.983	0.995	1.006	0.62	5.3	1.012	0.990	0.987	0.59	5.4	1.009	0.992	0.992
a lighthouse		1.4	2.2	0.985	0.989	0.980	1.8	2.6	0.995	0.988	0.986	1.8	2.6	0.995	0.988	0.988
an igloo		2.4	9.1	0.986	0.965	0.949	2.4	9.2	0.988	0.965	0.948	2.4	9.3	0.988	0.965	0.949
a log cabin	L	2.3	12	0.916	0.942	0.886	2.9	15	0.909	0.938	0.883	2.7	14	0.913	0.944	0.883
the Leaning Tower of Pisa	T, G, S	4.4	4.9	0.946	0.958	0.962	4.3	4.9	0.942	0.952	0.959	4.3	4.8	0.968	1.010	0.977
the Pantheon	L	2.3	2.5	0.998	0.981	0.937	2.3	2.5	1.002	0.978	0.951	2.2	2.4	1.001	0.978	0.944
Himeji Castle	G	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
a natural stone arch	G	9.0	11	0.686	0.798	0.700	9.1	11	0.685	0.796	0.692	9.0	11	0.687	0.796	0.694
the Taj Mahal		2.1	2.4	0.976	0.978	0.972	1.8	2.2	1.029	0.981	0.970	1.6	2.0	1.013	0.985	0.964
Median (all objects)		6.6	19	0.909	0.911	0.888	4.6	16	0.907	0.916	0.883	4.8	17	0.907	0.912	0.883
n		47	47	47	47	47	47	47	47	47	47	47	47	47	47	47
Median (unflagged)		3.4	20	0.928	0.929	0.895	3.0	14	0.940	0.940	0.899	2.7	17	0.941	0.942	0.899
n		18	18	18	18	18	18	18	18	18	18	18	18	18	18	18

Table S6: Computational cost at octree depth 7: retries and wall-clock times in seconds (GPU, after a warm-up) of the successful attempt; in parentheses, the total over all attempts where failed attempts add more than 0.5%. Solver retries: attempts after the first; mesh retries: failed TetGen calls plus rejected remeshes of the surface the kept mesh was built from (rejections inside unkept size-search trials not counted). Prep.: preparation (Nitsche: boundary conditions, SDF, octree, trimming, solver setup; P1: surface resizing, TetGen, boundary mapping, solver setup). Solve: Newton iteration to relative residual 10^{-6}; to 10^{-2}: the same solve until it first reaches 10^{-2}. Superscripts: ^{C}, CPU solve; ^{R}, reduced GPU solver; both not comparable with GPU times. Flags, at the session's $h_{\min}$: T, wall thinner than $4h_{\min}$; G, gaps narrower than $h_{\min}$ on over 1% of the surface; L, a load within $3h_{\min}$ of a support; S, a support piece below $50h_{\min}^2$ or a partial clamp on a section thinner than $4h_{\min}$; blank, unflagged. —, no body-fitted mesh (the input surface self-intersects). Medians over all objects and over the unflagged ones; n, number of values. Time medians are of successful attempts. Details in Section S**??**.

		Diffused Nitsche				Body-fitted P1														
						Matched DOFs					Matched surface					Graded				
Object	Flags	Solver retries	Prep. (s)	Solve (s)	To 10^{-2} (s)	Mesh retries	Solver retries	Prep. (s)	Solve (s)	To 10^{-2} (s)	Mesh retries	Solver retries	Prep. (s)	Solve (s)	To 10^{-2} (s)	Mesh retries	Solver retries	Prep. (s)	Solve (s)	To 10^{-2} (s)
a TV remote control		0	0.336	0.725	0.725	0	0	4.46	0.654	0.654	0	0	4.71	0.924	0.704	0	0	4.58	0.646	0.646
a USB stick		0	0.305	0.868	0.603	0	0	4.26	0.844	0.644	0	0	4.36	0.963	0.710	0	0	4.47	1.09	0.630
an electric toothbrush		0	0.172	0.939	0.785	0	0	3.63	2.17	1.44	0	0	3.55	1.60	1.15	0	0	3.83	1.87	1.50
a coffee mug	T	0	1.40	4.13	2.22	0	0	14.5	4.36	1.65	0	0	14.1	4.14	1.58	0	0	16.3	4.25	1.63
a computer mouse		0	0.585	0.731	0.411	0	0	6.67	0.800	0.525	0	0	11.4	1.50	0.950	0	0	7.39	1.08	0.747
a school chair with a steel frame and a plastic seat	T, S	0	0.491	3.60	2.51	0	1	6.55	2.96 (4.13)	1.79	0	1	5.78	2.54 (3.53)	1.52	0	1	9.40	4.55 (5.91)	2.29
a glass-top dining table	T, S	0	0.677	2.77	1.80	0	0	8.37	2.25	1.61	0	0	16.7	1.83	1.31	0	0	15.1	1.82	1.26
a park bench	T, S	0	0.695	3.40	2.27	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
a bar stool	S	0	0.498	2.64	1.72	0	0	6.47	2.92	1.77	0	1	5.42	1.93 (3.04)	1.34	0	1	7.54	2.73 (4.11)	1.88
a sofa	S	0	2.33	5.79	3.07	0	0	7.97	1.42	0.870	0	0	10.5	1.98	1.21	0	0	9.10	1.39	0.867
a teddy bear		0	0.603	3.08	1.89	0	0	7.30	2.82	2.30	0	0	11.4	4.37	3.57	0	0	8.26	2.81	2.30
a wooden rocking horse	T	0	0.573	1.42	1.42	0	0	6.79	2.17	1.51	0	0	5.79	1.58	1.12	0	0	7.87	2.18	1.51
a pogo stick	S	0	0.173	0.766	0.489	0	0	3.67	2.17	0.704	0	0	3.55	0.991	0.478	0	0	3.87	1.77	0.712
a wooden toy train engine	G, L, S	0	0.713	1.10	0.632	0	0	12.0	0.988	0.645	0	0	19.8	1.58	1.06	0	0	14.7	1.28	0.854
a yo-yo	G	0	1.70	3.10	1.63	0	0	18.4	4.67	1.93	0	0	33.8	10.3	7.27	0	0	19.3	3.15	1.85
a jelly dessert		0	2.16	6.21	5.25	0	0	25.1	2.21	1.68	0	0	58.3	8.80	7.05	0	0	17.5	3.98	3.25
a loaf of bread		0	1.17	24.5	1.52	0	0	13.2	11.5	1.62	0	0	36.2	40.1	5.47	0	0	15.1	11.4	1.60
a hamburger	G	0	1.18	21.8	1.53	0	0	13.0	11.8	1.58	0	0	34.7	43.9	5.33	0	0	14.9	11.7	1.58
a layered birthday cake		0	1.60	6.03	4.34	0	0	11.3	4.37	2.99	0	0	36.3	11.0	6.83	0	0	26.8	3.27	2.26
a slice of watermelon		0	0.643	1.41	0.749	0	0	6.86	1.02	0.577	0	0	11.6	2.00	1.08	0	0	7.95	0.992	0.563
a giraffe	S	0	0.260	1.26	0.865	0	0	4.22	1.58	1.12	0	0	4.08	1.35	0.956	0	0	4.65	2.08	1.14
an elephant	S	0	0.508	1.48	0.788	0	0	6.41	1.34	0.736	0	0	7.35	1.61	0.850	0	0	7.13	1.31	0.698
a rhinoceros	S	0	0.459	0.864	0.477	0	0	5.66	1.40	0.614	0	0	6.62	1.79	0.775	0	0	6.08	1.00	0.616
a horse	S	0	0.668	2.79	1.77	0	0	14.6	3.78	2.04	0	0	16.5	3.16	2.21	0	0	7.76	4.31	1.83
a crocodile	S	0	0.435	0.894	0.894	0	0	10.6	1.85	1.32	0	0	10.5	1.70	1.22	0	0	11.8	1.86	1.32
a car suspension control arm with rubber bushings		0	0.547	1.65	1.16	0	0	6.03	1.67	1.20	0	0	5.96	1.52	1.08	0	0	6.95	1.59	1.13

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Table S6: Computational cost at octree depth 7 (continued).

		Diffused Nitsche				Body-fitted P1														
						Matched DOFs					Matched surface					Graded				
Object	Flags	Solver retries	Prep. (s)	Solve (s)	To 10^{-2} (s)	Mesh retries	Solver retries	Prep. (s)	Solve (s)	To 10^{-2} (s)	Mesh retries	Solver retries	Prep. (s)	Solve (s)	To 10^{-2} (s)	Mesh retries	Solver retries	Prep. (s)	Solve (s)	To 10^{-2} (s)
a truck leaf spring		0	0.188	0.528	0.349	0	0	3.75	0.999	0.799	0	0	3.70	0.713	0.410	0	0	3.94	0.956	0.760
a car wheel with its rubber tyre	G	0	1.52	4.58	2.49	0	0	12.3	3.07	1.67	0	0	23.4	7.89	4.20	0	0	13.0	3.52	1.92
a car brake disc	S	0	1.02	1.76	0.987	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
a crankshaft	G	0	0.462	0.759	0.759	0	0	5.66	1.17	0.846	0	0	5.72	0.864	0.864	0	0	5.96	1.20	0.874
a road roller with a steel drum and rubber tyres	T, G, S	0	1.07	1.78	0.920	0	0	21.5	3.28	2.35	0	0	32.5	3.04	2.16	0	0	35.3	5.19	3.96
a jackhammer	S	0	0.200	2.04	0.795	0	3	3.97	2.46[C] (33.5)	1.70[C]	0	3	3.85	1.52[C] (22.9)	1.07[C]	0	4	4.22	2.48[C] (44.8)	1.71[C]
an industrial robot arm	G	0	0.537	1.66	1.10	0	0	6.78	2.23	1.23	0	0	6.93	2.37	1.31	0	0	7.65	2.16	1.19
an adjustable wrench	T, S	0	0.188	0.694	0.495	0	0	3.79	1.57	0.652	0	0	3.54	0.917	0.462	0	0	3.95	1.28	0.616
a claw hammer		0	0.227	1.08	0.565	0	0	3.91	1.18	0.670	0	0	3.78	0.993	0.589	0	0	4.17	1.52	0.739
a dental crown		0	1.55	2.47	1.30	0	0	21.9	3.10	2.03	0	3	76.1	148[C]	139[C]	0	0	18.5	2.21	1.36
a knee implant	G	0	1.49	3.18	1.61	0	0	14.3	3.07	1.76	0	0	26.0	6.92	3.92	0	0	12.0	2.79	1.60
a walking cane with a rubber tip	T, L, S	0	0.142	0.417	0.275	0	0	3.48	0.654	0.482	0	0	3.44	0.455	0.341	0	0	3.67	0.667	0.490
a human femur		0	0.257	1.08	0.659	0	0	4.22	1.57	0.904	0	0	4.11	1.65	0.851	0	0	4.71	1.89	0.943
a lumbar vertebra		0	0.897	1.42	0.709	0	0	16.8	1.04	0.634	0	0	41.4	1.55	0.955	0	0	7.80	0.991	0.603
a skyscraper	L	0	0.324	0.528	0.307	0	0	4.53	0.670	0.432	0	0	5.28	0.834	0.523	0	0	4.88	0.650	0.421
a thatched stone cottage	L	0	1.08	2.43	1.29	0	0	11.7	1.72	0.947	0	0	30.3	5.48	2.94	0	0	13.0	1.67	0.928
a lighthouse		0	0.671	0.979	0.584	0	0	7.31	1.00	0.615	0	0	10.8	1.59	0.984	0	0	8.38	1.06	0.659
an igloo		0	1.41	1.84	0.943	0	0	12.1	1.68	1.05	0	0	13.9	1.95	1.21	0	0	13.7	1.62	1.01
a log cabin	L	0	1.29	1.83	0.977	0	0	14.5	1.95	1.20	0	0	34.3	6.09	3.63	0	0	16.1	1.88	1.15
the Leaning Tower of Pisa	T, G, S	0	0.593	0.993	0.555	2	1	22.2	8.48 (10.4)	3.21	2	1	24.6	6.40 (8.66)	1.64	8	1	21.6 (48.4)	4.89 (6.53)	1.99
the Pantheon	L	0	1.06	1.61	0.891	0	0	11.3	1.50	0.957	0	0	13.4	1.80	1.14	0	0	12.8	1.44	0.915
Himeji Castle	G	0	1.52	3.82	2.04	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
a natural stone arch	G	0	1.45	2.05	1.15	0	0	33.0	2.15	0.970	0	0	36.7	5.99	3.10	0	0	12.4	1.75	1.10
the Taj Mahal		0	1.03	2.55	1.36	0	0	11.7	2.03	1.13	0	0	18.8	3.85	2.15	0	0	13.7	1.88	1.05
Median (all objects)		0	0.655	1.71	0.960	0	0	7.31	1.95	1.13	0	0	11.4	1.83	1.21	0	0	8.26	1.86	1.13
n		50	50	50	50	47	47	47	47	47	47	47	47	47	47	47	47	47	47	47
Median (unflagged)		0	0.623	1.42	0.767	0	0	7.08	1.62	0.975	0	0	11.4	1.62	1.08	0	0	7.87	1.60	0.976
n		18	18	18	18	18	18	18	18	18	18	18	18	18	18	18	18	18	18	18

Table S7: Degrees of freedom (in thousands) of diffused Nitsche (three per free node of the trimmed octree) and of the three body-fitted P1 meshes (three per vertex) at octree depth 8. Matched DOFs: vertex count targeted at the octree's node count (secant search, closest of up to six trials kept); matched surface: surface edge length $h_{\min}$; graded: $h_{\min}$ at the surface, coarser inside like the octree. Flags, at the session's $h_{\min}$: T, wall thinner than $4h_{\min}$; G, gaps narrower than $h_{\min}$ on over 1% of the surface; L, a load within $3h_{\min}$ of a support; S, a support piece below $50h_{\min}^2$ or a partial clamp on a section thinner than $4h_{\min}$; blank, unflagged. —, no body-fitted mesh (the input surface self-intersects). Medians over all objects and over the unflagged ones; n, number of values. Details in Section S**??**.

			Body-fitted P1						Body-fitted P1		
Object	Flags	Diffused Nitsche	Matched DOFs	Matched surface	Graded	Object	Flags	Diffused Nitsche	Matched DOFs	Matched surface	Graded
a TV remote control		209	210	443	208	a car suspension control arm with rubber bushings		353	358	588	350
a USB stick		193	193	378	193	a truck leaf spring		58.0	56.8	54.3	57.6
an electric toothbrush		66.7	68.1	74.3	65.8	a car wheel with its rubber tyre	G	1410	1460	3580	1390
a coffee mug	T	1260	1270	1970	1250	a car brake disc	S	912	—	—	—
a computer mouse		445	446	1890	443	a crankshaft	G	299	309	552	296
a school chair with a steel frame and a plastic seat	T, S	336	344	453	335	a road roller with a steel drum and rubber tyres	T, G, S	454	464	501	437
a glass-top dining table	T, S	293	217	193	293	a jackhammer	S	87.6	90.8	105	84.2
a park bench	T, S	400	—	—	—	an industrial robot arm	G	398	402	723	392
a bar stool	S	353	351	348	348	an adjustable wrench	T, S	76.9	44.9	55.2	76.4
a sofa	S	544	569	1510	541	a claw hammer		113	119	144	111
a teddy bear		486	453	1550	482	a dental crown		1480	1470	9040	1470
a wooden rocking horse	T	412	423	476	407	a knee implant	G	1310	1280	4950	1300
a pogo stick	S	48.0	47.0	30.6	46.6	a walking cane with a rubber tip	T, L, S	30.6	29.2	18.2	30.2
a wooden toy train engine	G, L, S	438	455	1700	454	a human femur		153	155	214	150
a yo-yo	G	1560	1510	4490	1550	a lumbar vertebra		418	423	1130	415
a jelly dessert		831	829	4890	826	a skyscraper	L	218	207	620	216
a loaf of bread		983	984	6730	979	a thatched stone cottage	L	897	902	5710	894
a hamburger	G	1000	999	6690	997	a lighthouse		437	434	1480	443
a layered birthday cake		841	846	4520	840	an igloo		1070	1080	2470	1070
a slice of watermelon		473	460	2000	477	a log cabin	L	1130	1140	6420	1120
a giraffe	S	145	152	181	141	the Leaning Tower of Pisa	T, G, S	422	433	709	341
an elephant	S	404	412	1010	399	the Pantheon	L	935	968	2190	965
a rhinoceros	S	333	351	904	330	Himeji Castle	G	1420	—	—	—
a horse	S	354	358	761	349	a natural stone arch	G	825	835	2790	822
a crocodile	S	169	170	244	167	the Taj Mahal		880	862	3320	902
Median (all objects)		420	423	904	407	Median (unflagged)		441	440	1510	443
n		50	47	47	47	n		18	18	18	18

Table S8: Discrepancy of diffused Nitsche from each body-fitted P1 solution at octree depth 8. η_{rig} and $\eta_{\text{rig}}^{H^1}$: relative L^2 and H^1 norms of the displacement difference after removing its best-fit rigid motion (in %); $\ell_{\text{N}}/\ell_{\text{P1}}$: compliance ratio (below 1: Nitsche stiffer); s_{95}, s_{99}: Nitsche-to-P1 ratios of the 95th and 99th percentiles of the von Mises stress. Flags, at the session's $h_{\min}$: T, wall thinner than $4h_{\min}$; G, gaps narrower than $h_{\min}$ on over 1% of the surface; L, a load within $3h_{\min}$ of a support; S, a support piece below $50h_{\min}^2$ or a partial clamp on a section thinner than $4h_{\min}$; blank, unflagged. Gray rows: η_{rig} above 10% against at least one mesh. —, no body-fitted mesh (the input surface self-intersects). † Nitsche solve converged only to 10^{-2}; excluded from the medians. Medians over all objects and over the unflagged ones; n, number of values. Details in Section S**??**.

		Body-fitted P1														
		Matched DOFs					Matched surface					Graded				
Object	Flags	η_{rig} (%)	$\eta_{\text{rig}}^{H^1}$ (%)	$\ell_{\text{N}}/\ell_{\text{P1}}$	s_{95}	s_{99}	η_{rig} (%)	$\eta_{\text{rig}}^{H^1}$ (%)	$\ell_{\text{N}}/\ell_{\text{P1}}$	s_{95}	s_{99}	η_{rig} (%)	$\eta_{\text{rig}}^{H^1}$ (%)	$\ell_{\text{N}}/\ell_{\text{P1}}$	s_{95}	s_{99}
a TV remote control		2.6	5.1	0.929	0.948	0.940	2.8	5.3	0.925	0.942	0.933	2.4	4.8	0.934	0.950	0.949
a USB stick		0.43	2.0	0.938	0.965	0.924	0.49	2.1	0.934	0.963	0.915	0.33	1.9	0.949	0.972	0.931
an electric toothbrush		6.1	10	1.091	0.950	1.014	5.2	8.5	1.077	0.943	1.025	2.9	4.9	1.042	0.942	0.977
a coffee mug	T	1.7	4.3	0.943	0.954	0.937	2.0	4.7	0.938	0.951	0.933	1.8	4.3	0.944	0.952	0.938
a computer mouse		3.4	16	0.975	0.997	0.970	3.1	14	0.976	0.999	0.957	2.8	15	0.980	1.003	0.960
a school chair with a steel frame and a plastic seat	T, S	4.2	5.7	0.932	0.934	0.917	5.5	7.5	0.908	0.917	0.902	4.3	5.9	0.926	0.926	0.917
a glass-top dining table	T, S	6.2	11	0.951	0.975	0.929	5.1	9.2	0.960	0.981	0.932	10	18	0.921	0.962	0.917
a park bench	T, S	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
a bar stool	S	5.3	10	0.949	0.954	0.933	5.2	10	0.949	0.954	0.931	5.6	11	0.945	0.951	0.926
a sofa	S	0.58	5.3	0.997	0.973	0.963	0.71	5.3	0.991	0.966	0.960	0.49	5.2	0.998	0.981	0.964
a teddy bear		0.58	1.8	0.926	0.936	0.958	0.60	1.8	0.929	0.985	0.951	0.55	1.7	0.935	0.988	0.948
a wooden rocking horse	T	2.7	5.3	0.960	0.966	0.969	2.9	5.6	0.956	0.965	0.967	2.7	5.3	0.960	0.966	0.969
a pogo stick	S	16	46	0.957	0.876	1.042	16	46	0.965	0.886	1.047	16	46	0.957	0.881	1.042
a wooden toy train engine	G, L, S	11	40	0.901	0.971	0.874	10	41	0.905	0.975	0.869	9.8	40	0.914	0.978	0.876
a yo-yo	G	16	48	0.606	0.761	0.674	16	48	0.596	0.751	0.663	16	48	0.601	0.754	0.667
a jelly dessert		1.8	18	0.976	0.971	0.972	2.2	17	0.974	0.960	0.955	1.6	18	0.982	0.976	0.973
a loaf of bread†		2.0	16	0.972	0.998	0.965	2.7	16	0.935	0.956	0.927	2.3	17	0.943	0.966	0.930
a hamburger	G	5.3	19	0.939	0.963	0.917	5.6	19	0.933	0.954	0.905	3.8	19	0.955	0.973	0.935
a layered birthday cake		1.4	13	0.990	0.981	0.994	1.6	14	0.991	0.974	0.987	1.3	15	0.992	0.976	0.986
a slice of watermelon		2.7	15	0.947	0.968	0.935	1.9	14	0.960	0.971	0.933	1.5	15	0.967	0.978	0.938
a giraffe	S	0.84	3.0	1.048	0.988	1.034	0.89	3.1	1.051	0.988	1.032	0.87	2.9	1.032	0.984	1.029
an elephant	S	0.42	2.5	0.993	0.997	1.003	0.49	2.6	0.994	0.997	1.002	0.46	2.7	0.996	0.998	1.002
a rhinoceros	S	0.67	4.8	0.991	0.984	1.005	0.75	4.7	0.992	0.997	1.003	0.70	4.8	0.993	0.993	1.003
a horse	S	26	54	0.967	0.968	0.981	26	54	0.949	0.966	0.969	26	54	0.951	0.966	0.969
a crocodile	S	1.9	5.6	0.905	0.901	0.912	2.3	5.9	0.891	0.894	0.898	2.2	5.9	0.891	0.901	0.891
a car suspension control arm with rubber bushings		4.2	11	0.968	0.974	0.979	4.1	11	0.964	0.972	0.975	4.2	11	0.969	0.975	0.980
a truck leaf spring		14	24	0.831	0.856	0.727	14	24	0.832	0.857	0.729	15	25	0.824	0.859	0.718
a car wheel with its rubber tyre	G	2.8	7.5	0.924	0.950	0.925	3.1	8.1	0.918	0.946	0.917	2.9	7.9	0.925	0.950	0.923
a car brake disc	S	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
a crankshaft	G	5.4	14	0.769	0.797	0.753	4.7	12	0.801	0.820	0.783	4.6	12	0.806	0.824	0.785
a road roller with a steel drum and rubber tyres	T, G, S	37	42	0.493	0.654	0.592	37	42	0.491	0.655	0.592	37	42	0.489	0.653	0.589
a jackhammer	S	10	16	0.841	0.983	0.836	11	16	0.834	0.984	0.829	11	17	0.831	0.984	0.820
an industrial robot arm	G	20	46	0.556	0.737	0.607	20	47	0.549	0.733	0.600	20	47	0.553	0.736	0.605
an adjustable wrench	T, S	0.27	2.2	1.007	1.008	1.002	0.27	2.2	1.007	1.006	1.004	0.31	2.3	1.005	1.004	1.001

Continued on next page

Table S8: Discrepancy at octree depth 8 (continued).

		Body-fitted P1														
		Matched DOFs					Matched surface					Graded				
Object	Flags	η_{rig} (%)	$\eta_{\mathrm{rig}}^{H^1}$ (%)	$\ell_{\mathrm{N}}/\ell_{\mathrm{P1}}$	s_{95}	s_{99}	η_{rig} (%)	$\eta_{\mathrm{rig}}^{H^1}$ (%)	$\ell_{\mathrm{N}}/\ell_{\mathrm{P1}}$	s_{95}	s_{99}	η_{rig} (%)	$\eta_{\mathrm{rig}}^{H^1}$ (%)	$\ell_{\mathrm{N}}/\ell_{\mathrm{P1}}$	s_{95}	s_{99}
a claw hammer		0.42	1.8	0.951	0.955	0.968	0.45	1.9	0.949	0.953	0.962	0.39	1.7	0.953	0.955	0.971
a dental crown		4.5	16	0.966	0.920	0.886	4.6	16	0.967	0.917	0.884	4.5	16	0.968	0.920	0.884
a knee implant	G	65	80	0.456	0.487	0.428	65	81	0.451	0.480	0.422	65	80	0.458	0.486	0.427
a walking cane with a rubber tip	T, L, S	7.6	13	0.958	0.899	0.836	6.4	11	0.967	0.913	0.868	9.0	15	0.937	0.889	0.830
a human femur		0.18	0.94	0.992	0.989	0.981	0.26	1.0	0.990	0.985	0.979	0.10	0.89	0.995	0.992	0.983
a lumbar vertebra		4.1	16	0.961	0.941	0.934	3.9	16	0.965	0.939	0.933	3.8	16	0.966	0.941	0.934
a skyscraper	L	0.13	0.16	0.994	0.994	0.992	0.10	0.14	1.003	0.993	0.990	0.081	0.13	1.003	0.992	0.994
a thatched stone cottage	L	0.70	4.1	0.985	0.984	0.982	0.53	4.3	0.998	0.986	0.975	0.47	4.4	0.997	0.987	0.976
a lighthouse		0.90	1.4	0.994	0.997	0.991	1.6	2.2	0.996	0.991	0.992	1.3	1.9	0.996	0.992	0.992
an igloo		2.5	7.9	0.971	0.965	0.952	1.5	6.4	0.991	0.981	0.977	1.5	6.8	0.989	0.981	0.977
a log cabin	L	1.5	8.6	0.945	0.954	0.965	2.0	11	0.933	0.954	0.936	1.8	11	0.939	0.960	0.935
the Leaning Tower of Pisa	T, G, S	3.7	4.2	0.964	0.961	0.958	3.7	4.2	0.965	0.958	0.956	4.6	5.4	0.969	0.965	0.969
the Pantheon	L	0.96	1.1	0.992	0.984	0.973	1.2	1.3	1.004	0.991	0.981	0.96	1.1	1.002	0.988	0.985
Himeji Castle	G	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
a natural stone arch	G	6.7	8.3	0.776	0.867	0.797	6.7	8.4	0.773	0.864	0.792	6.7	8.3	0.776	0.865	0.796
the Taj Mahal		1.2	1.4	0.980	0.987	0.989	0.95	1.3	1.013	0.992	0.989	0.77	1.2	1.003	0.996	0.990
Median (all objects)		2.7	8.4	0.958	0.964	0.955	3.0	8.4	0.960	0.961	0.944	2.8	8.1	0.956	0.966	0.948
n		46	46	46	46	46	46	46	46	46	46	46	46	46	46	46
Median (unflagged)		2.5	10	0.968	0.965	0.968	1.9	8.5	0.967	0.971	0.957	1.5	6.8	0.969	0.976	0.971
n		17	17	17	17	17	17	17	17	17	17	17	17	17	17	17

Table S9: Computational cost at octree depth 8: retries and wall-clock times in seconds (GPU, after a warm-up) of the successful attempt; in parentheses, the total over all attempts where failed attempts add more than 0.5%. Solver retries: attempts after the first; mesh retries: failed TetGen calls plus rejected remeshes of the surface the kept mesh was built from (rejections inside unkept size-search trials not counted). Prep.: preparation (Nitsche: boundary conditions, SDF, octree, trimming, solver setup; P1: surface resizing, TetGen, boundary mapping, solver setup). Solve: Newton iteration to relative residual 10^{-6}; to 10^{-2}: the same solve until it first reaches 10^{-2}. Superscripts: ^{C}, CPU solve; ^{R}, reduced GPU solver; both not comparable with GPU times; †, converged only to 10^{-2}, excluded from the median. Flags, at the session's $h_{\min}$: T, wall thinner than $4h_{\min}$; G, gaps narrower than $h_{\min}$ on over 1% of the surface; L, a load within $3h_{\min}$ of a support; S, a support piece below $50h_{\min}^2$ or a partial clamp on a section thinner than $4h_{\min}$; blank, unflagged. —, no body-fitted mesh (the input surface self-intersects). Medians over all objects and over the unflagged ones; n, number of values. Time medians are of successful attempts. Details in Section S**??**.

		Diffused Nitsche				Body-fitted P1														
						Matched DOFs					Matched surface					Graded				
Object	Flags	Solver retries	Prep. (s)	Solve (s)	To 10^{-2} (s)	Mesh retries	Solver retries	Prep. (s)	Solve (s)	To 10^{-2} (s)	Mesh retries	Solver retries	Prep. (s)	Solve (s)	To 10^{-2} (s)	Mesh retries	Solver retries	Prep. (s)	Solve (s)	To 10^{-2} (s)
a TV remote control		0	1.15	2.66	2.66	0	0	8.74	2.63	1.96	0	0	17.2	5.58	4.07	0	0	9.77	2.56	1.90
a USB stick		0	1.08	3.97	2.02	0	0	8.01	2.69	1.91	0	0	15.0	6.96	3.75	0	0	9.17	3.44	1.87
an electric toothbrush		0	0.505	3.22	2.17	0	0	4.29	5.26	3.62	0	0	4.49	4.53	3.17	0	0	4.68	9.73	8.05
a coffee mug	T	0	6.25	29.8	22.1	0	0	73.0	24.4	17.5	0	0	77.4	127^{C}	114^{C}	0	0	80.1	23.8	16.9
a computer mouse		0	2.36	2.86	1.51	0	0	18.7	2.75	1.71	0	0	72.3	437^{C}	423^{C}	0	0	55.6	3.58	2.41
a school chair with a steel frame and a plastic seat	T, S	1	1.79 (3.52)	14.1 (17.6)	7.12	0	1	10.9	11.5 (14.5)	6.94	0	1	14.2	20.6 (24.5)	8.96	0	1	38.5	13.7 (16.7)	6.90
a glass-top dining table	T, S	0	1.55	8.85	5.90	0	1	7.89	4.16 (6.31)	2.75	0	1	6.75	4.42 (6.48)	2.86	0	1	33.3	6.10 (8.82)	4.09
a park bench	T, S	1	2.38 (4.78)	12.4 (16.6)	8.28	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
a bar stool	S	1	1.78 (3.52)	10.4 (14.0)	6.73	0	1	14.5	7.91 (11.3)	5.30	0	1	14.2	7.81 (11.2)	5.23	0	1	24.6	10.4 (13.7)	5.24
a sofa	S	0	2.64	9.56	3.51	0	0	25.2	6.31	3.74	0	0	58.4	158^{C}	147^{C}	0	0	30.5	19.6	10.8
a teddy bear		0	2.36	12.7	7.67	0	0	22.3	12.1	9.81	0	0	59.8	1743^{C}	1436^{C}	0	0	37.1	13.4	10.9
a wooden rocking horse	T	0	2.10	6.55	6.55	0	1	16.5	10.2 (14.3)	6.93	0	1	18.7	11.1 (15.7)	7.55	0	1	27.0	9.56 (13.4)	6.60
a pogo stick	S	1	0.400 (0.793)	3.20 (5.77)	0.748	0	1	3.79	5.98 (7.12)	0.800	0	0	3.50	1.97	0.620	0	1	4.32	4.79 (5.90)	0.792
a wooden toy train engine	G, L, S	0	2.95	4.40	2.40	0	0	44.2	4.20	2.68	0	0	248	263^{C}	245^{C}	0	0	29.8	6.07	4.10
a yo-yo	G	0	7.45	15.4	7.92	0	0	137	270^{C}	248^{C}	0	0	181	874^{C}	824^{C}	0	0	161	101^{C}	91.2^{C}
a jelly dessert		0	3.97	12.6	9.61	0	0	54.7	11.7	9.51	0	1	378	71.5^{R} (2284)	50.1^{R}	0	0	75.6	10.1	6.23
a loaf of bread		6	4.74 (32.6)	$1991^{\mathrm{C}\dagger}$ (8765)	170^{C}	0	0	40.2	47.4	6.32	0	1	281	365^{R} (2549)	40.7^{R}	0	0	53.4	59.9	7.17
a hamburger	G	0	4.96	110	6.10	0	0	88.5	53.3	6.53	0	1	347	338^{R} (1473)	38.8^{R}	0	0	51.0	51.5	6.26
a layered birthday cake		0	3.80	16.1	10.3	0	0	67.6	9.95	6.15	0	1	408	67.7^{R} (1259)	36.4^{R}	0	0	82.2	9.36	5.82
a slice of watermelon		0	2.59	5.44	5.44	0	0	35.5	3.77	2.01	2	0	164 (472)	911^{C}	885^{C}	0	0	96.3	8.66	6.99
a giraffe	S	0	0.945	4.56	3.20	0	1	6.94	6.27 (8.20)	2.53	0	1	8.15	7.31 (9.42)	2.95	0	1	10.4	8.48 (10.3)	3.15

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Table S9: Computational cost at octree depth 8 (continued).

		Diffused Nitsche				Body-fitted P1														
						Matched DOFs					Matched surface					Graded				
Object	Flags	Solver retries	Prep. (s)	Solve (s)	To 10^{-2} (s)	Mesh retries	Solver retries	Prep. (s)	Solve (s)	To 10^{-2} (s)	Mesh retries	Solver retries	Prep. (s)	Solve (s)	To 10^{-2} (s)	Mesh retries	Solver retries	Prep. (s)	Solve (s)	To 10^{-2} (s)
an elephant	S	0	1.99	6.39	3.20	0	0	15.4	7.72	2.76	0	0	38.6	21.2	15.2	0	0	22.3	7.60	2.69
a rhinoceros	S	0	1.71	3.63	1.90	0	0	13.6	5.49	3.83	0	0	34.2	15.3	10.6	0	0	17.1	5.16	2.08
a horse	S	1	1.86 (3.74)	21.0 (32.1)	7.20	0	1	13.9	13.4 (16.8)	5.42	0	1	68.6	51.6 (59.0)	11.5	0	1	22.1	18.3 (21.7)	5.27
a crocodile	S	0	1.18	5.45	3.74	0	1	8.80	6.37 (8.86)	3.49	0	1	25.2	7.84 (11.2)	3.96	0	1	11.4	6.03 (8.06)	3.13
a car suspension control arm with rubber bushings		0	1.90	7.12	5.13	0	0	13.6	10.5	4.62	0	0	22.7	18.0	7.91	0	0	25.6	8.33	4.53
a truck leaf spring		0	0.452	1.89	1.43	0	0	3.85	1.94	1.51	0	0	3.83	1.91	1.47	0	0	4.19	2.02	1.57
a car wheel with its rubber tyre	G	0	6.78	23.1	11.7	0	0	125	244[C]	210[C]	0	0	409	558[C]	500[C]	0	0	68.2	57.1[C]	44.0[C]
a car brake disc	S	0	4.85	9.57	5.31	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
a crankshaft	G	0	1.57	4.82	3.29	0	0	13.5	4.57	3.22	0	0	19.7	8.41	5.86	0	0	11.9	4.20	2.96
a road roller with a steel drum and rubber tyres	T, G, S	0	3.66	15.4	4.85	0	1	16.8	13.3 (17.8)	6.67	0	1	65.2	10.6 (17.2)	7.18	0	1	38.1	9.14 (13.1)	5.97
a jackhammer	S	1	0.571 (1.15)	34.2 (35.5)	25.9	0	3	4.77	9.35[C] (48.4)	6.25[C]	0	3	5.12	11.3[C] (49.4)	7.61[C]	0	3	6.46	8.39[C] (56.4)	5.61[C]
an industrial robot arm	G	0	2.08	7.78	5.34	0	0	15.0	11.8	5.09	0	0	28.1	22.7	14.0	0	0	27.0	9.17	4.87
an adjustable wrench	T, S	0	0.573	2.71	1.45	0	0	3.81	1.84	1.02	0	0	4.02	4.45	1.63	0	0	5.26	3.26	1.46
a claw hammer		0	0.739	3.34	1.76	0	0	5.81	3.82	2.39	0	0	6.69	4.49	1.98	0	0	7.12	3.62	1.62
a dental crown		0	6.58	9.94	5.15	0	0	53.1	193[C]	182[C]	0	0	387	1496[C]	1088[C]	0	0	72.1	71.0[C]	62.0[C]
a knee implant	G	0	5.96	15.2	7.41	0	3	46.8	114[C]	105[C]	0	0	190	794[C]	575[C]	0	0	76.6	14.2	7.83
a walking cane with a rubber tip	T, L, S	0	0.315	0.947	0.621	0	0	3.56	1.24	0.855	0	0	3.42	0.969	0.686	0	0	3.81	1.25	0.872
a human femur		0	0.871	4.00	2.37	0	0	6.96	4.05	1.91	0	0	9.19	5.56	3.61	0	0	9.19	3.99	2.57
a lumbar vertebra		0	2.17	5.10	1.83	0	0	16.1	5.19	3.73	0	0	43.5	14.8	5.98	0	0	31.3	5.05	3.63
a skyscraper	L	0	1.12	1.95	1.05	0	0	9.90	1.75	1.07	0	0	23.8	5.34	3.12	0	0	10.3	1.84	1.13
a thatched stone cottage	L	0	4.16	9.93	5.12	0	0	36.5	6.89	3.69	0	0	232	2796[C]	1487[C]	0	0	46.5	6.53	3.50
a lighthouse		0	2.19	4.20	2.24	0	0	43.0	4.76	2.92	0	0	57.6	158[C]	148[C]	0	0	23.3	5.06	2.97
an igloo		0	5.45	7.61	3.97	0	0	67.6	9.33	6.11	0	0	129	271[C]	243[C]	0	0	72.6	6.88	4.14
a log cabin	L	0	5.40	8.28	4.39	0	0	51.7	8.79	5.22	0	0	265	1759[C]	1178[C]	0	0	56.7	10.3	5.94
the Leaning Tower of Pisa	T, G, S	0	2.17	4.73	2.62	0	1	15.2	12.1 (16.1)	3.00	0	1	26.4	20.3 (26.7)	10.2	8	1	13.5 (82.2)	21.1 (24.0)	4.62
the Pantheon	L	0	4.46	7.60	4.23	0	0	35.3	8.60	5.63	0	0	78.1	225[C]	210[C]	0	0	86.4	7.87	4.82

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Table S9: Computational cost at octree depth 8 (continued).

		Diffused Nitsche				Body-fitted P1															
						Matched DOFs					Matched surface					Graded					
Object	Flags	Solver retries	Prep. (s)	Solve (s)	To 10^{-2} (s)	Mesh retries	Solver retries	Prep. (s)	Solve (s)	To 10^{-2} (s)	Mesh retries	Solver retries	Prep. (s)	Solve (s)	To 10^{-2} (s)	Mesh retries	Solver retries	Prep. (s)	Solve (s)	To 10^{-2} (s)
Himeji Castle	G	0	6.76	19.7	10.7	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
a natural stone arch	G	0	4.47	7.31	3.89	1	0	43.2	9.53	6.77	0	0	155	490[C]	470[C]	0	0	93.4	7.18	4.42
the Taj Mahal		0	3.93	11.3	6.04	0	0	162	11.2	5.72	0	0	124	1228[C]	1164[C]	0	0	269	14.6	7.86
Median (all objects)		0	2.18	7.60	4.62	0	0	16.1	7.91	3.83	0	0	43.5	21.2	11.5	0	0	29.8	8.39	4.62
n		50	50	49	50	47	47	47	47	47	47	47	47	47	47	47	47	47	47	47
Median (unflagged)		0	2.28	5.10	3.31	0	0	20.5	5.23	3.68	0	0	58.7	69.6	38.5	0	0	34.2	7.60	4.34
n		18	18	17	18	18	18	18	18	18	18	18	18	18	18	18	18	18	18	18